SHIBAURA INSTITUTE OF TECHNOLOGY

Curring and Uncurrying Transformation with Eclipse IDE

Luiz Gustavo M. Sampaio
z09017@sic.shibaura-it.ac.jp

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What is Currying?

- Turn **one function** with **multiples arguments** in **multiples functions** with **one argument**

Function return

- fun div(x, y) = x / y;
*val div = fn : real * real -> real*

- div(3.0, 2.0);
val it = 1.5 : real

Function domain

- fun div x y = x / y;
val div = fn : real -> real -> real

- div 3.0;
val it = fn : real -> real

- it 2.0;
val it = 1.5 : real

Currying

Uncurrying

Currying and recursion

- Really works!

```
- fun times n x =
    if n = 0 then 1 else x * times (n-1) x;
val times = fn : int -> ( int -> int )

- val fiveTimes =  times 5;
val fiveTimes = fn : int -> int

- fiveTimes 2;
val it = 32 : int

- fiveTimes 7;
val it = 16807 : int
```

And so on…

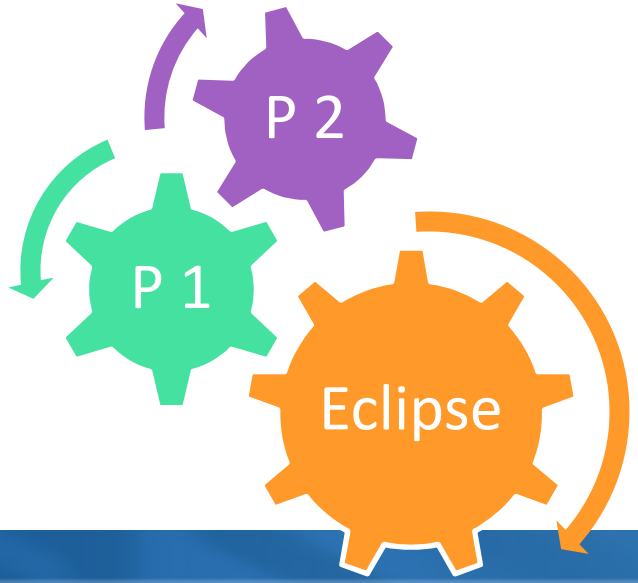
Plug-in development: Why Eclipse?

- Multi-language platform
- Widely used around the world
- Highly extensible
- Great documentation on Internet

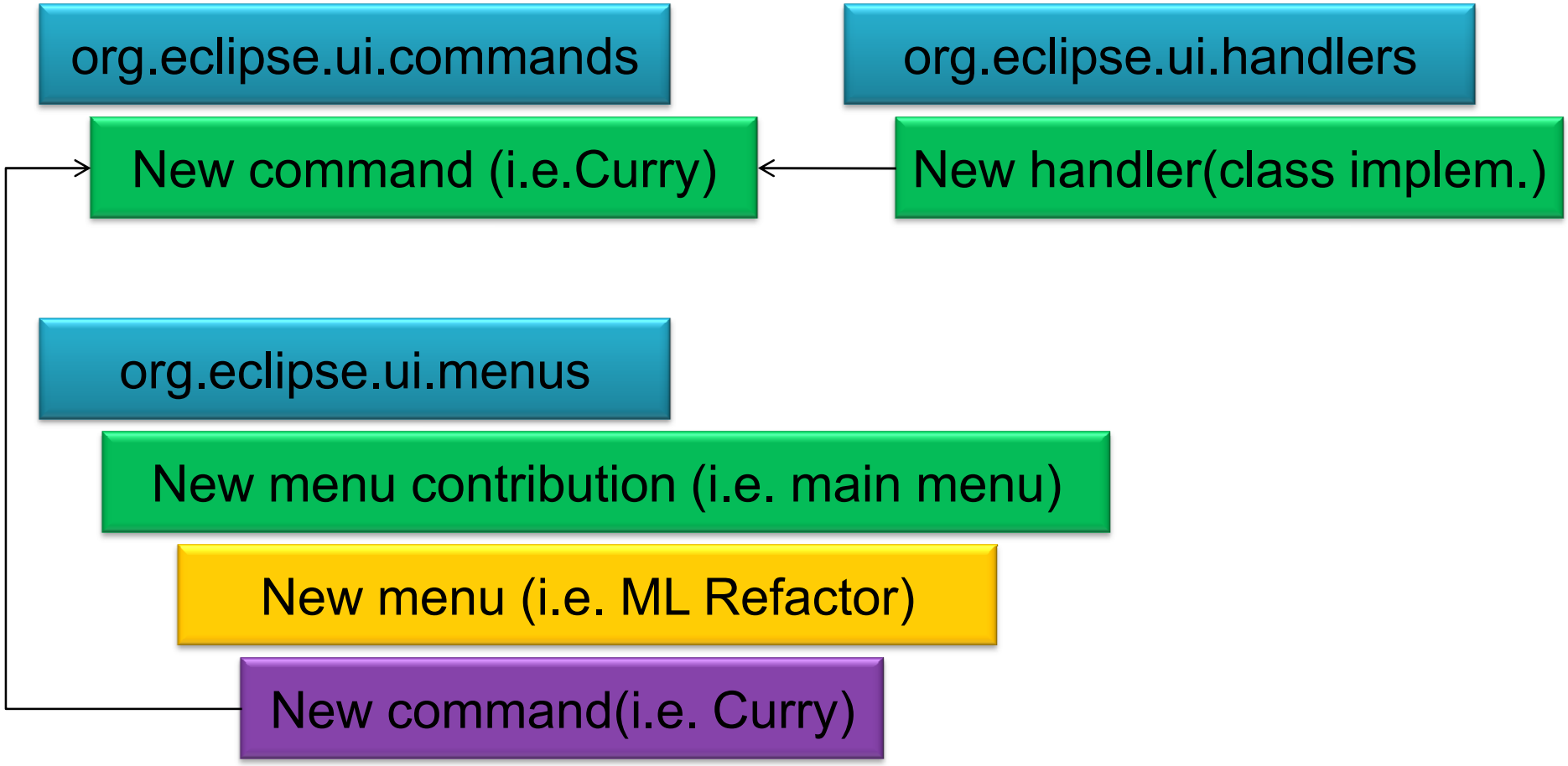
Plug-in development: Eclipse APIs

- Extension Point: Defines the possibility to add functionality (i.e. menus, windows, toolbars)
- When developing a plug-in you can provide your owns extension points, to be expanded by others.

How use the extension points and other details will be explained step-by-step in our wiki soon



Example: How add a menu item



Currying on Eclipse: How it will be?

- User open a file with Standard ML code
- Select the function to be curried
- Acces the menu *ML Refactor > Curry*
- Plug-in read the file and parse
- If the function syntax is “OK”, apply the curry

How is the state of the plug-in now?

- Menu implementation – *ok*
- Read the SML file – *ok*
- JFlex (Lexer generator for Java) – *ok*
- CUP (Parser generator for Java) – *requires more studying*
- Grammar for SML – *requires more studying*
- How to apply the currying/uncurrying in the file – *requires more studying*

Demonstration